

Exp. 7-Restricting Data Using Group By Clause

Objective:At the end of this experiment, participants will understand how to summarise and analyse data using GROUP BY and aggregate functions, enabling meaningful insights, grouped reporting, and comparison across categories.

Total and average salaries for employees in a chosen job:

Query1:

```
SELECT SUM(salary) AS total_salary,  
       AVG(salary) AS avg_salary,  
       COUNT(*) AS emp_count  
FROM hr.employees  
WHERE job_id = 'SA_REP';
```

Output1:

Download ▾ Execution time: 0.007 seconds			
	TOTAL_SALARY	AVG_SALARY	EMP_COUNT
1	250500	8350	30

Total and average salaries grouped by payment-type analogue:

Query2:

```
SELECT job_id,  
       COUNT(*) AS emp_count,  
       SUM(salary) AS total_salary,  
       ROUND(AVG(salary),2) AS avg_salary  
FROM hr.employees  
GROUP BY job_id  
ORDER BY total_salary DESC;
```

Output2:

Download ▾ Execution time: 0.007 seconds				
	JOB_ID	EMP_COUNT	TOTAL_SALARY	AVG_SALARY
1	SA_REP	30	250500	8350
2	SH_CLERK	20	64300	3215
3	SA_MAN	5	61000	12200
4	ST_CLERK	20	55700	2785
5	FI_ACCOUNT	5	39600	7920
6	ST_MAN	5	36400	7280
7	AD_VP	2	34000	17000
8	IT_PROG	5	28800	5760

How many distinct hire_dates compared to total employees:

Query3:

```
SELECT COUNT(DISTINCT hire_date) AS distinct_hire_dates,  
       COUNT(*) AS total_employees  
FROM hr.employees;
```

Output3:

		Download ▾	Execution time: 0.006 seconds
	DISTINCT_HIRE_DA	TOTAL_EMPLOYEES	
1	98	107	

How many distinct managers and departments versus total employees:

Query4:

```
SELECT COUNT(DISTINCT manager_id) AS distinct_managers,  
       COUNT(DISTINCT department_id) AS distinct_departments,  
       COUNT(*) AS total_employees  
FROM hr.employees;
```

Output4:

		Download ▾	Execution time: 0.006 seconds
	DISTINCT_MANAGE	DISTINCT_DEPARTM	TOTAL_EMPLOYEES
1	18	11	107

Lowest and highest salaries among employees:

Query5:

```
SELECT MIN(salary) AS lowest_salary,  
       MAX(salary) AS highest_salary  
FROM hr.employees;
```

Output5:

		Download ▾	Execution time: 0.007 seconds
	LOWEST_SALARY	HIGHEST_SALARY	
1	2100	24000	